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Usability Test

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Executive Summary

This study evaluates the usability of the USDA Agricultural Commodity Import Requirements (ACIR) portal through moderated, task-based usability testing. ACIR provides import requirements for over 100,000 country and commodity combinations. It serves importers, customs brokers, Customs and Border Protection officers, and APHIS staff.

Before testing with participants, the team ran one internal pilot with the client to refine the script and tasks. The pilot revealed that one task needed clearer wording and confirmed that sessions fit within the 30 to 45 minute target. The team then conducted seven moderated tests with regulatory professionals who held varying levels of ACIR experience. Sessions ran over Zoom, and participants shared their screens so the team could observe navigation in real time. Participants thought aloud while completing three tasks. The first task asked them to determine the import status of a fresh apricot from Mexico for human consumption. The second asked them to find Small Lots of Seed guidance in the Plants for Planting Manual and connect it to the matching CIR document. The third asked them to verify a scientific name and find admissibility and quarantine information. A pre-test and post-test questionnaire captured background, confidence, and reactions. Each session was logged by at least one notetaker, and every team member moderated at least one session.

Three findings stood out across the tests. Search and filter use worked well when participants entered the exact scientific name and tile, but small input mismatches and overlapping result sets caused friction. The link between manual guidance and the CIR document was rarely clear, which led to long detours through PDFs and unrelated pages. The Taxonomy page, which resolves scientific name questions, was hard to discover from the main search flow.

The team recommends three priority improvements. First, ACIR should support more forgiving search behavior with stronger autocomplete and synonym matching so users can recover from minor input errors. Second, the portal should add direct links between manual sections and their matching CIR documents so users can move between sources without relying on prior knowledge. Third, the Taxonomy page should be easier to reach from the main search flow, especially when a query returns no exact match.

Introduction

The ACIR portal is a public USDA APHIS resource that helps users find the regulations that apply when bringing plants and plant products into the United States. The system covers over 100,000 country and commodity combinations and supports importers, customs brokers, Customs and Border Protection officers, and APHIS staff. Public content makes up roughly 99 percent of portal traffic. The core task is simple in concept. A user enters a country and a commodity, then reads the matching Commodity Import Requirement document to confirm admissibility and any permit, certificate, or treatment rules.

Reference manuals like the Plants for Planting Manual sit alongside the CIRs for deeper regulatory guidance, and a separate Taxonomy page handles scientific name lookups.

This usability test extends earlier findings from the team's interviews, comparative evaluation, and heuristic evaluation. The team focused on three questions. Can a person familiar with the domain reach the correct import decision using ACIR alone? Where does the workflow break down between the search results, the CIR document, and the supporting manual content? How do users handle scientific names that do not match exactly, such as synonyms and renamed taxa? Usability testing fits these questions because the answers depend on observable behavior under realistic task pressure. Surveys and interviews cannot capture where users hesitate, backtrack, or accept the wrong answer. The findings give the client direct evidence to prioritize fixes.

Methods

Participants

The target population was regulatory professionals who look up or interpret commodity import requirements as part of their work. Because ACIR serves a restricted government user base, recruitment was handled by the client, who invited professionals across departments and locations. The team specified that participants should vary in job function, experience level, and portal familiarity. Seven moderated tests were conducted. Participants included a national policy manager with nearly 20 years of import/export experience, a private-sector purchasing specialist who imports from roughly 40 to 50 countries, a training specialist who instructs APHIS officers on ACIR, a CBP agriculture specialist with over 20 years of field experience who prefers legacy PDF manuals, an import program manager who uses ACIR three to four times per week, and additional regulatory specialists with varying levels of portal familiarity. This range ensured the study captured both expert and less-experienced perspectives.

Pilot Testing

Before testing with participants, the team ran one internal pilot with the client contact to verify task clarity and session timing. The pilot revealed that Task 2 needed more specific wording because the original prompt left it unclear whether participants should start in the manual or the main portal. The team revised the language to explicitly name the Small Lots of Seed pathway and separate the manual lookup from the CIR identification step. The pilot also confirmed that sessions fit within 30 to 45 minutes including pre-test questions and debriefing.

Setting and Protocol

All sessions were conducted remotely over Zoom. Participants joined from their own workstations and shared their screens so the team could observe navigation in real time. Sessions were video-recorded with consent. Each session followed four phases: preamble, pre-test questions, task performance, and

debriefing. The moderator opened by explaining that the test evaluated the portal, not the participant, and asked participants to think aloud throughout. Three pre-test questions gathered each participant's background with import regulations, prior ACIR use, and comfort with policy documentation. Each team member moderated at least one session, and each test was logged by at least one dedicated notetaker who recorded timestamps, navigation paths, errors, and verbal indicators of confusion.

Tasks

Three tasks were designed around the usability issues most consistently identified in prior research phases. Task 1 asked participants to determine the import status of fresh *Prunus armeniaca* from Mexico for human consumption, evaluating the core search-to-result workflow. Task 2 asked participants to locate the Small Lots of Seed section in the Plants for Planting Manual for *Solanum lycopersicum* seeds from Italy and connect it to the corresponding CIR document, targeting the manual-to-CIR cross-referencing gap flagged in interviews and the heuristic evaluation. Task 3 asked participants to verify whether *Leucanthemum nipponicum* was the accepted scientific name, find the correct import requirement for the plant from Dominica, and determine quarantine status, testing synonym resolution and the Taxonomy page's discoverability. Tasks progressed in complexity to build participant comfort before introducing harder scenarios.

Post-Test Questionnaire

After all tasks, the moderator asked five closing questions covering which task felt easiest and hardest, whether any labels or messages were unclear, whether participants ever felt unsure they had the right document, and what single change would most improve ACIR.

Findings/Recommendations

Across the usability test transcripts, several clear and consistent findings emerged regarding how users interact with the ACIR portal. Overall, findings are described in a way that multiple audiences can understand, as they focus on observable behaviors (navigation paths, confusion points, and decision-making cues) rather than technical jargon. For example, participants consistently relied on entering a commodity and country to reach a Commodity Import Requirement (CIR) page, which indicates that the core workflow is generally intuitive for experienced users.

However, a key finding is that information gaps and unclear system connections disrupt otherwise successful workflows. Participants frequently identified missing or unclear links, such as references to “related documents” that were not actually present, forcing them to search elsewhere or rely on prior knowledge. Additionally, users struggled to locate supporting information like pest-free area lists, indicating breakdowns between primary results and secondary resources.

Another major finding is that system usability varies significantly by user expertise. Experienced users reported that determining import status was “easy” when they understood domain-specific inputs like plant parts or scientific names, but less experienced or differently specialized users took inefficient paths, such as navigating PDFs or unrelated manuals. This highlights a reliance on prior knowledge rather than system guidance.

The findings are supported with strong behavioral evidence, including think-aloud commentary, navigation decisions, and explicit statements of confusion or confidence. Confidence levels were generally high when users reached CIR pages with clear requirements, but dropped when encountering ambiguous labels (e.g., lack of explicit “permitted/restricted” status) or missing links.

Overall, the findings are presented with appropriate confidence: patterns are consistent across participants, but clearly tied to observed behaviors rather than overgeneralized claims.

Based on these findings, we recommend three priority improvements. First, ACIR should support more forgiving search behavior, including autocomplete, synonym matching, and common-name support, to help users recover from minor input mismatches and scientific-name uncertainty. Second, the portal should provide direct links between manual guidance and the relevant CIR documents so users can move between sources without relying on prior knowledge or lengthy PDF searches. Third, the Taxonomy page should be surfaced more clearly within the main search flow, especially when a user’s query returns no exact match, so users can more easily resolve synonym and accepted-name issues. Together, these changes would reduce navigation friction, improve interpretive confidence, and support both expert and less-experienced users

Discussion

While the study provided high-quality data from expert users, the findings are limited by specific constraints in the participant pool. Because recruitment was handled by the client, the participant selection was restricted to professionals already within the existing government ecosystem, which likely excluded the unique pain points and mental models of external or completely new users. Furthermore, while the range of expertise captured was broad, the small sample size (n=7) means that some specialized edge cases in regulatory workflows may not have been fully observed.

To build on these findings, the next phase of the project should focus on high-fidelity prototyping of the recommended “Permitted/Prohibited” status banners to evaluate improvements in user speed and confidence. Additionally, a longitudinal study or a more extensive A/B test is suggested to determine if improved hyperlinking between manuals and CIR documents successfully reduces the “taxing” cognitive load observed during Task 2.

Appendices

Appendix A: Recruitment message

The recruitment process was managed by the client, who invited a diverse group of regulatory professionals across various departments based on the team's specific requirements for a variety of job functions, experience levels, and degrees of portal familiarity.

[Link to email regarding correspondence with client](#)

Appendix B: Moderator Script

Opening Script:

Hi, my name is [Name], and I'm part of a student team from the University of Michigan School of Information. and thank you for joining today.

We're testing the ACIR manuals and portal, not you. There are no right or wrong answers here. We want to understand how easy or difficult it is to find information, interpret what the system is telling you, and complete common tasks.

As you work, please think aloud. Tell me what you're looking for, what you expect to happen, what feels confusing, and how confident you are in your answers.

I may ask follow-up questions, but I will not guide you through the tasks unless you get completely stuck.

Do you have any questions before we begin?

Pre-Test Questions

- 1. Can you briefly describe your experience with import regulations, inspection systems, or plant-related documentation?*
- 2. Have you used ACIR or a similar regulatory portal before?*
- 3. How comfortable are you with navigating large manuals or policy documentation?*

(Task 1: Determine Import Status for a Commodity)

Moderator Reads:

Imagine an importer wants to bring fresh Prunus armeniaca from Mexico into the United States for human consumption. They ask whether this commodity is allowed to enter and what requirements apply.

Using the ACIR portal, determine:

- *whether the commodity is prohibited, restricted, or permitted, and*
- *what documentation, certification, or inspection requirements apply for importing *Prunus armeniaca* for consumption.*

Please talk through how you reach your answer.

Please begin whenever you're ready.

What this evaluates

- Ability to use ACIR search and filtering features
- Interpretation of admissibility results
- Understanding of status language such as prohibited, restricted, or permitted
- Ability to locate supporting import requirements
- Confidence making a decision under time pressure

Related findings

- Finding 2 (inconsistent regulatory language)
- Finding 4 (search/navigation friction)

Moderator Notes to Capture

- Where the participant starts
- Whether they use search, filters, or browse by category
- Whether they correctly identify **fresh** and **human consumption** as relevant task constraints
- Whether they understand the admissibility result immediately
- Whether they confuse commodity status with documentation requirements
- Whether they open multiple pages or tiles to verify the answer
- Time to first relevant result
- Signs of hesitation, uncertainty, or backtracking

Intended Answers:

- Import Requirement: *Apricot (Fruit) from Mexico into All Ports*
- Admissible, Import Permit Required, Must Have a Phytosanitary Certificate

Follow-Up Questions

- *What did the status label (definition needed) mean to you?*
- *Was it easy to tell whether the commodity was allowed to enter?*
- *What helped you decide you had the right answer?*

- *Was the language around requirements clear or confusing?*
- *Did you feel confident in your final answer? Why or why not?*

(Task 2: Locate Regulatory Guidance in the Manual)

Moderator Reads:

*Imagine you are helping someone import *Solanum lycopersicum* seeds from Italy into the United States under the Small Lots of Seed pathway. You need to find the manual guidance and connect it to the relevant CIR document.*

*Using the ACIR portal, locate the section in the Plants for Planting Manual that discusses Small Lots of Seed requirements for *Solanum lycopersicum* seeds.*

Next, identify the CIR Document associated with these requirements when imported from Italy and determine the relevant regulatory guidance that explains what documentation, restrictions, or certifications are required.

Please begin whenever you're ready.

What this evaluates

- Ability to move between ACIR and manual-based guidance
- Efficiency of search and navigation
- Ability to connect commodity results to CIR documentation
- Scannability of manual content and hyperlinks
- Interpretive confidence when information is spread across multiple sources

Related findings

- Finding 1 (scannability issues with hyperlink manuals)
- Finding 2 (inconsistent regulatory language)
- Finding 4 (search/navigation friction)

Moderator Notes to Capture

- Where the participant begins
- Whether they search by commodity, country, or manual term
- Whether they recognize that **Small Lots of Seed** is the relevant pathway
- Whether they use hyperlinks, table of contents, in-page search, or scanning
- Time to first relevant section
- Number of wrong turns or backtracks
- Whether they successfully connect the manual guidance to the CIR document

- Visible confusion about document names, labels, or navigation structure
- Statements showing overload or uncertainty

Intended Answers:

- Manual page: Plants for Planting - Seeds (There is a Small Lots of Seed section)
- Import Requirement: Inadmissible Small Lots of Seed of Solanum lycopersicum from All Countries
- Inadmissible

Follow-Up Questions

- *How easy or difficult was it to find the correct section of the manual?*
- *What helped you know you were in the right place?*
- *Was it clear how the CIR Document connected to the manual guidance?*
- *Did the hyperlinks help you or make the task harder?*
- *How confident are you that the requirements you found are correct?*

(Task 3: Verify Commodity Classification and Synonym Handling)

Moderator Reads:

Imagine you are assisting an importer who wants to bring Leucanthemum nipponicum (loocanthamum newponicome) plants from Dominica into the United States for planting. Before giving guidance, you want to confirm whether that scientific name is accepted in ACIR and determine the plant's admissibility requirements.

Using the ACIR portal, determine:

- *whether Leucanthemum nipponicum is the accepted scientific name and, if not, what the accepted name is,*
- *whether you can locate the appropriate import requirement document for the plant (not the seed),*
- *whether the plant is admissible, and*
- *whether it can enter without being quarantined.*

Please explain what information you are using to make each decision.

What this evaluates

- Ability to handle scientific synonyms and taxonomy issues
- Cross-referencing between taxonomy, commodity, and requirement records
- Ability to distinguish **plant** from **seed** pathways
- Understanding of admissibility and quarantine-related requirements

- Trust in system outputs when names or classifications do not match exactly

Related findings

- Finding 3 (contradictory classification data)
- Finding 4 (search/navigation friction)
- Finding 2 (inconsistent regulatory language)

Moderator Notes to Capture

- Whether the participant searches using the provided scientific name first
- Whether they notice or recover from synonym issues
- Whether they understand that the task is about the **plant**, not seed
- How they determine the accepted scientific name
- Whether they can locate the correct import requirement document
- Whether they interpret admissibility and quarantine language correctly
- Whether they cross-check multiple pages, tiles, or documents
- Signs of confusion, second-guessing, or mistrust of the result

Intended Answers:

- *Leucanthemum nipponicum* is not the accepted scientific name
- *Nipponanthemum nipponicum* is the accepted name (Best outcome is users find this on the [Taxonomy page](#) but interested in where they go to get this info)
- Import Requirement: [Postentry Quarantine Admissible Herbaceous Plants for Planting, Nipponanthemum nipponicum from Non-PCN Countries](#)
- Admissible, but the plant must be quarantined to enter (Important Information)

Follow-Up Questions

- *What made you think the scientific name was or was not accepted?*
- *How easy was it to recover when the original name did not match exactly?*
- *Was it clear which result applied to the plant rather than the seed?*
- *How did you determine whether the plant could enter without quarantine?*
- *How confident are you in the answer you found?*

Closing Questions

- *Which task felt the easiest, and why?*
- *Which task felt the most confusing or frustrating?*
- *Were there any labels, terms, or status messages that were unclear?*
- *Did you ever feel unsure whether you were looking at the right document or result?*

- *What would make ACIR easier to use for these kinds of import questions?*

Closing Script:

Thank you for completing the session. Your feedback is very helpful and will be used to improve the ACIR manuals and portal experience.

Before we wrap up, is there anything else you want to share about your experience using the system today?

Appendix C: Summaries of pre and post questionnaire responses

Test #1: Customs and Border Protection (CBP) Non-Propagative

- **Pre-questionnaire summary:** The participant has extensive experience with legacy paper and PDF manuals. While they have successfully transitioned to the ACIR portal and use it daily, they still feel more "comfortable" with the precision of the older systems compared to newer officers who only know ACIR.
- **Post-questionnaire summary:** Task 1 was the easiest due to familiar search parameters. The most frustrating part was the "multi-step" feel of manual navigation. They noted that while they are confident, the system needs to be more intuitive for "novice" users who lack a background in the old manuals.

Test #2: Quarantine Policy, Analysis, and Support (QPAS) Non-Propagative

- **Pre-questionnaire summary:** A 20-year veteran of regulatory policy with a high level of expertise in import/export documentation. Extremely comfortable with complex manuals but expressed skepticism about digital "gaps" in the transition.
- **Post-questionnaire summary:** Task 1 was easiest because it aligned with their standard workflow. Task 3 was the most frustrating due to "family-level" search results that required excessive reading. Recommended clear status banners (Prohibited/Allowed) to improve efficiency.

Test #3: Seed Importer Plants for Planting

- **Pre-questionnaire summary:** An experienced "super user" with 6 years in the field. Regularly navigates complex seed-related regulations and maintains a high comfort level with technical documentation.
- **Post-questionnaire summary:** Task 2 was the easiest due to professional familiarity with seed content. Task 3 was the most frustrating because it "failed" at undefined technical jargon (like "PCN country"). Suggested adding inline tooltips for specialized terms.

Test #4: Center for Agricultural Safeguarding Training All Areas

- **Pre-questionnaire summary:** A training specialist with 3 years of experience teaching USDA APHIS officers. Very comfortable with the system's logic and filtering capabilities as part of their instructional role.
- **Post-questionnaire summary:** Task 1 was the easiest as the search criteria were clearly defined. Task 3 was the most difficult due to the "live" nature of the database (documents archiving mid-session) and the complexity of synonym lookups.

Test #5: Plant Inspection Station (PIS) Plants for Planting

- **Pre-questionnaire summary:** A long-term veteran (since 2001) who explicitly prefers searching PDF manuals over the digital portal interface. They find the portal more "taxing" than traditional methods.
- **Post-questionnaire summary:** Found Task 1 the most straightforward. Task 3 felt "overwhelming" and "overloaded" with clicks. They recommended that the search functionality should be more "Google-like" to handle synonyms and common names more gracefully.

Test #6: Plant Inspection Station (PIS) Plants for Planting

- **Pre-questionnaire summary:** An Import Program Manager who uses ACIR 3–4 times a week to manage bilateral work plans and audit programs. Highly comfortable with regulatory documentation.
- **Post-questionnaire summary:** Task 1 was easiest. Task 2 was frustrating because "Small Lots of Seed" was difficult to find within the manual hierarchy without prior knowledge. Suggested that labels and status messages are currently too wordy for quick interpretation.

Test #7: Preclearance and Offshore Programs Non-Propagative

- **Pre-questionnaire summary:** Participant works with a vast amount of internal regulatory information and uses the system frequently. High comfort level with navigating large-scale policy databases.
- **Post-questionnaire summary:** Task 1 was the most efficient. Task 3 was successful but highlighted the friction of scientific name requirements. Recommended that the system bridge the gap between common names and scientific names to help less-specialized users.

Appendix D: Summaries of each individual test

Test #1: Customs and Border Protection (CBP) Non-Propagative

- **Moderator:** Layilah Campbell, Micaela Ciambrone

- **Pre-Test:** Participant has extensive experience with older manual systems but has transitioned to using the ACIR portal.
- **Task 1:** Successfully determined that fresh apricots from Mexico are admissible with an import permit and phytosanitary certificate. Found it straightforward but cautioned about the "intended use" filter.
- **Task 2:** Navigated to the "Plants for Planting" manual and found the "Small Lots of Seed" section. Identified the CIR document but noted the multi-step process could be confusing for novices.
- **Task 3:** Initially found no results for the scientific name. Used the taxonomy tool to find the accepted name (*Nipponanthemum nipponicum*) and correctly identified the requirement for postentry quarantine.

Test #2: Quarantine Policy, Analysis, and Support (QPAS) Non-Propagative

- **Moderator:** Dongsub Kim, Seorin Jang
- **Pre-Test:** 20-year veteran of regulatory policy.
- **Task 1:** Found the navigation functional but noted the lack of explicit "admissibility" status labels, which required reading through multiple results to confirm status.
- **Task 2:** Identified a critical gap where manual cross-references led to missing or unavailable documents (e.g., "Designated Pest-Free Areas").
- **Task 3:** Rated as the most difficult task. Struggled with the system hierarchy pulling results at the family level, which cluttered the search with irrelevant results.

Test #3: Seed Importer Plants for Planting

- **Moderator:** Dongsub Kim, Shenghao Mei
- **Pre-Test:** A "super user" with approximately 6 years of experience.
- **Task 1:** Completed successfully with high confidence once the correct section was located.
- **Task 2:** Found difficult; required searching multiple disconnected documents with no clear navigational pathway between the manual and CIRs.
- **Task 3: Incomplete.** The participant could not find or confirm information due to undefined terminology (e.g., "PCN country") and inability to verify if the species name was accepted.

Test #4: Center for Agricultural Safeguarding Training All Areas

- **Moderator:** Dongsub Kim, Shenghao Mei
- **Pre-Test:** 3 years of experience instructing USDA APHIS officers.
- **Task 1:** Easiest task because all required search components were provided upfront.
- **Task 2:** Successfully navigated, but noted that precise scenario descriptions (e.g., fresh vs. cut fruit) are essential to avoid selecting the wrong CIR.

- **Task 3:** Most difficult task due to the combination of synonym lookup, navigating archived documentation, and distinguishing between genus and species-level regulations.

Test #5: Plant Inspection Station (PIS) Plants for Planting

- **Moderator:** Seorin Jang, Layilah Campbell
- **Pre-Test:** Experience dating back to 2001; prefers PDF-based manuals over the digital portal.
- **Task 1:** Located requirements but found ACIR filters less intuitive than a standard PDF search.
- **Task 2:** Described manual navigation as "taxing" and had difficulty connecting the manual section to the specific CIR document.
- **Task 3:** Struggled with the synonym lookup and felt "overloaded" by the number of steps required to reach a final answer.

Test #6: Plant Inspection Station (PIS) Plants for Planting

- **Moderator:** Micaela Ciambrone, Shenghao Mei
- **Pre-Test:** Manages import programs with bilateral work plans; uses ACIR 3-4 times a week.
- **Task 1:** Identified status quickly but suggested a clear "Permitted/Prohibited" banner would improve speed.
- **Task 2:** Found searching by "Small Lots of Seed" difficult without knowing exactly where the term sat in the manual hierarchy.
- **Task 3:** Friction with the taxonomy search; noted that users without scientific backgrounds would likely struggle to recover from synonym issues.

Test #7: Preclearance and Offshore Programs Non-Propagative

- **Moderator:** Micaela Ciambrone, Seorin Jang
- **Pre-Test:** Works with a vast amount of internal regulatory information.
- **Task 1:** Efficiently completed, though noted that broad keywords can sometimes pull "irrelevant responses."
- **Task 2:** Successfully navigated the manual but found the "Small Lots of Seed" section lacked direct hyperlinks to the expected CIRs.
- **Task 3:** Successfully used the taxonomy tab to find the synonym; suggested mapping common names to scientific names to reduce complexity for general users.

Appendix E: Basic Task Performance Data

Test #	Task 1 Status	Task 1 Time	Task 2 Status	Task 2 Time	Task 3 Status	Task 3 Time
1	Success	3:42	Success	4:55	Success	5:32

2	Success	2:25	Success	1:38	Success	2:03
3	Success	5:47	Success	11:52	Incomplete	N/A
4	Success	4:12	Success	5:45	Success	8:15
5	Success	5:36	Success	2:46	Success	4:29
6	Success	5:53	Success	4:23	Success	3:53
7	Success	3:15	Success	5:10	Success	6:25

Appendix F: Debrief notes (Refined)

Test #1: Customs and Border Protection (CBP) Non-Propagative

- Participant noted that while they are confident, the system needs to be more intuitive for "novice" users who lack a background in the old manuals.
- Observed that the "multi-step" process of navigating manuals and identifying CIR documents could be confusing for those not transitioned from legacy paper or PDF systems.

Test #2: Quarantine Policy, Analysis, and Support (QPAS) Non-Propagative

- Participant identified a critical gap where manual cross-references led to missing or unavailable documents, specifically "Designated Pest-Free Areas".
- Recommended the implementation of clear "Prohibited/Allowed" status banners to improve the efficiency of identifying admissibility.

Test #3: Seed Importer Plants for Planting

- Participant noted that the system "failed" at undefined technical jargon, citing "PCN country" as a specific example.
- Suggested adding inline tooltips for specialized terms to reduce the need for external verification.

Test #4: Center for Agricultural Safeguarding Training All Areas

- Observed confusion regarding the "live" nature of the database, specifically noting that documents appeared to be archiving mid-session.
- Emphasized that precise scenario descriptions (e.g., fresh vs. cut fruit) are essential to avoid selecting the wrong CIR.

Test #5: Plant Inspection Station (PIS) Plants for Planting

- Participant described manual navigation as "taxing" and felt "overloaded" and "overwhelmed" by the number of clicks required.
- Recommended that the search functionality should be more "Google-like" to handle synonyms and common names more gracefully.

Test #6: Plant Inspection Station (PIS) Plants for Planting

- Observed that "Small Lots of Seed" was difficult to find within the manual hierarchy without prior knowledge of where the term sat.
- Suggested that current labels and status messages are too wordy for quick interpretation.

Test #7: Preclearance and Offshore Programs Non-Propagative

- Highlighted friction regarding scientific name requirements and the fact that broad keywords can pull "irrelevant responses".
- Recommended mapping common names to scientific names to reduce complexity for general users.

Appendix G: Data logging form

Task 1: Determine Import Status for a Commodity

- **Start Time / End Time:**
- **Navigation Path:** Recorded whether the participant used the main search bar, side filters, or browsed by category.
- **Admissibility Identification:** Logged whether the user correctly identified the status of fresh apricots from Mexico for human consumption.
- **Search Errors:** Captured instances where broad keywords pulled "irrelevant responses" or where "intended use" filters caused confusion.
- **Task Outcome:** Marked as "Success" if the participant correctly identified the need for an import permit and phytosanitary certificate.

Task 2: Locate Regulatory Guidance in the Manual

- **Start Time / End Time:**
- **Entry Point:** Noted if the participant started in the ACIR portal search or navigated directly to the "Plants for Planting" manual.
- **Pathway Tracking:** Recorded the ability to locate the specific "Small Lots of Seed" section within the manual hierarchy.
- **Cross-Referencing Success:** Logged whether the participant could successfully connect the manual guidance back to the corresponding CIR document.
- **Observation Notes:** Documented "taxing" navigation or frustration due to missing hyperlinks and disconnected documents.

Task 3: Verify Commodity Classification and Synonym Handling

- **Start Time / End Time:**
- **Search Input:** Logged the initial scientific name used and the participant's reaction when no exact match was found.
- **Taxonomy Tool Use:** Recorded whether the participant discovered and correctly used the Taxonomy page to find the accepted name, *Nipponanthemum nipponicum*.

- **Interpretation of Results:** Documented whether the user identified the postentry quarantine requirement for the plant.
- **Clutter/Friction Notes:** Logged instances where irrelevant "family-level" results or undefined terms like "PCN country" caused hesitation or incomplete tasks.